Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Course: B. Tech
Code: 102602

Subject: Dynamics of Machinery

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.
(v) Engineering drawing instruments and drawing sheets permitted.

Q.1 ***Choose the correct option / answer the following (Any seven question only):*** **[2 x 7 = 14]**

(a) In an automobile, if the vehicle makes a left turn, the gyroscope torque
(i) Increases the forces on outer wheels
(ii) Decreases the forces on outer wheels
(iii) Does not affect the forces on the outer wheels
(iv) None of the above

(b) The factor which affects the critical speed of a shaft is
(i) Diameter of the disc
(ii) Span of the shaft
(iii) Eccentricity
(iv) All of these

(c) The size of a cam depends upon
(i) Base circle
(ii) Pitch circle
(iii) Prime circle
(iv) Pitch curve

(d) The ratio of the maximum fluctuation of speed to the mean speed is called
(i) Fluctuation of speed
(ii) Maximum fluctuation of speed
(iii) Coefficient of fluctuation of speed
(iv) None of these

(e) The equation of free vibrations of a system is $1\ddot{x} + 36\pi^2 = 0$. Its natural frequency is
(i) 6 Hz
(ii) 3π Hz
(iii) 3 Hz
(iv) 6π Hz

(f) Offset is provided to a cam follower mechanism to
(i) Minimise the side thrust
(ii) Avoid jerk
(iii) accelerate
(iv) None of these

(g) In a slider crank mechanism, at which crank angle the connecting rod has zero angular velocity?
(i) 0°
(ii) 90°
(iii) 45°
(iv) None of these

(h) The follower normally used in aircraft engine is
(i) Roller
(ii) Flat faced
(iii) Knife edged
(iv) None of these

(i) The natural frequency of free torsional vibrations of a shaft is
(i) $2\pi\sqrt{\frac{q}{I}}$
(ii) $2\pi\sqrt{q.I}$
(iii) $\frac{1}{2\pi}\sqrt{\frac{q}{I}}$
(iv) $\frac{1}{2\pi}\sqrt{q.I}$

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Q1) In an engine, the work done by inertia forces in a cycle is

(i) Positive (iii) Negative

(ii) Zero (iv) None of these

Q.2 (a) Draw the turning moment diagram of a single cylinder double acting steam engine. [7]

(b) The mass of flywheel of an engine is 6.5 tones and the radius of gyration is 1.8 meters. It is found from the turning moment diagram that the fluctuation of energy is 56 KN-m. if the mean speed of the engine is 120 rpm, find the maximum and minimum speeds. [7]

Q.3 (a) How are cams classified? Described in detail. [5]

(b) Draw the profile a cam operating a knife-edge follower having a lift of 30 mm. the cam raises the follower with SHM for 150° of the rotation followed by a period of dwell for 60°. The follower descends for the next 100° rotation of the cam with uniform velocity, again followed by a dwell period. The cam rotates at a uniform velocity of 120 rpm and has a least radius of 20 mm. what will be the maximum velocity and acceleration of the follower during the lift and the return? [9]

Q.4 The following data relate to an uncoupled locomotive with cranks at 90°. [14]
Wheel center distance = 1.30 m, Cylinder center distance = 0.55 m, Diameter of driving wheel tread = 1.6 m, Radius of balance mass = 0.72 m, Crank radius = 0.32 m.
Mass of reciprocating parts per cylinder = 280 kg, Mass of revolving parts per cylinder = 230 kg.
Find:

(i) The magnitude and position of balancing masses in the planes of the driving wheels to balance all revolving masses and two-thirds of reciprocating masses.

(ii) The hammer blow at a speed of 105 km/h.

Q.5 (a) What do you mean by 'balancing of reciprocating engine'? [7]

(b) Explain the terms: Primary disturbing force and secondary disturbing force. [7]

Q.6 (a) What do you mean by gyroscopic couple? Derive a relation for its magnitude. [7]

(b) A uniform disc having a mass of 8 kg and a radius of gyration of 150 mm mounted on one end of a horizontal arm of 200 mm length. The other end of the arm can rotate freely in a universal bearing. The disc is given a clockwise spin of 250 rpm as seen from the disc end of the arm. Determine the motion of the disc if the arm remains horizontal. [7]

Q.7 (a) Explain the term 'whirling speed' or 'critical speed'. [7]

(b) The measurements on a mechanical vibrating system show that it has a mass of 8 kg and that the springs can be combined to give an equivalent spring of stiffness 5.4 N/mm. If the vibrating system have a dashpot attached which exerts a force of 40 N when the mass has a velocity of 1 m/s, find: 1. Critical damping coefficient. 2. Damping factor. 3. Logarithmic decrement. [7]

Q.8 (a) Explain the effects of partial balancing of locomotives. [7]

(b) State and explain D' Alembert's principle applied for the dynamic force analysis of links mechanisms. [7]

Q.9 (a) What do you understand by transmissibility? [7]

(b) In a spring-mass system, the mass is 0.1 kg and the stiffness of the spring is 1 kN/m. By introducing a damper, the frequency of oscillation is found to be 90% of the original value. What is the damping coefficient of the damper? [7]

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